

Chemical Manufacturing Analytics Project

End-to-End SQL, Excel, and Power BI Analysis

Project Report

A portfolio analysis of sales performance, customer value, manufacturing efficiency, supplier quality, and inventory movement using a simulated chemical manufacturing dataset.

Component	Tool / Deliverable
Database & analysis	MySQL / SQL
Validation & EDA	SQL
Visualization	Microsoft Excel
Interactive dashboard	Microsoft Power BI
Documentation	README + ERD + SQL scripts

1. Executive Summary

This project analyzes a simulated chemical manufacturing business using a relational database containing customer, order, product, production, inventory, supplier, raw-material, and quality-test data. The objective was to demonstrate an end-to-end analytics workflow: validate the data, use SQL to answer business questions, interpret results, and communicate findings through Excel and Power BI.

Six business questions were selected for the final analysis. Two additional questions were evaluated but excluded because the dataset produced limited differentiation: customer product variety was tightly clustered, while every order had the same one-day order-to-ship interval.

Key findings

- Product revenue was relatively evenly distributed, with no single product dominating the portfolio.
- Top-revenue customers generated value mainly through purchasing frequency rather than unusually high average order values.
- October recorded the highest order volume and revenue, while September and April were among the weakest months.
- Laguna Manufacturing Plant demonstrated the strongest production cost efficiency, with the lowest cost per kilogram despite the highest production volume.
- Nexus Raw Materials had the highest supplier-associated defect rate at 7.14%, although the overall supplier range was relatively narrow.
- Slip Additive SA-12, Copper Sulfate Pentahydrate, and Acetone had the highest inventory movement; several products also showed meaningful net inventory increases or decreases.

Overall conclusion

The analysis suggests that management should prioritize balanced portfolio management, customer retention, manufacturing-efficiency benchmarking, supplier-quality monitoring, and closer inventory monitoring. Findings are descriptive signals rather than causal conclusions because the dataset does not include every operational driver.

2. Project Overview

Objective

The objective was to use a relational chemical-manufacturing dataset to demonstrate practical data analytics skills and produce business-oriented insights rather than only technical SQL outputs.

Business context

The simulated company manufactures and sells chemical products to customers across multiple industries. Its operations involve raw-material suppliers, manufacturing plants, production batches, quality testing, inventory transactions, customer orders, and shipments.

Analytical workflow

- Relational database setup and schema design.
- Data validation for completeness, duplicates, consistency, and relationship integrity.
- Exploratory analysis performed alongside the business-question investigation.
- SQL-based aggregation and KPI calculation.
- Interpretation of findings into business insights and recommendations.
- Excel visualization as an analytical communication layer.
- Power BI dashboard development for interactive visualization.
- Documentation through README, ERD, SQL scripts, and this report.

3. Dataset and Data Model

The database is organized as a relational schema with connected entities representing the major business processes. Primary and foreign-key relationships connect transactions to customers, products, suppliers, plants, production batches, and quality tests.

Core tables

Table	Purpose
customers	Customer/company information and customer tier
orders	Customer order dates and customer relationships
order_items	Products, quantities, and unit prices within orders
products	Product names, categories, units, and selling prices
shipments	Shipment dates, delivery dates, and freight costs
inventory_transactions	Inbound and outbound inventory movements
plants	Manufacturing plant information
production_batches	Production volume and production cost by batch
raw_materials	Raw materials, suppliers, and unit costs
suppliers	Supplier information and supplier ratings
batch_material_usage	Raw-material usage within production batches
quality_tests	Batch quality results and defect information

Data validation

The dataset was validated for missing values, duplicates, inconsistent text values, date relationships, and table-linking issues. Import and relationship issues identified during setup were corrected before final analysis. The project emphasizes validation rather than presenting a separate data-cleaning script.

4. Methodology

SQL analysis

MySQL was used as the primary analytical environment. Queries used joins, aggregation, conditional logic, window functions, ranking, distinct counts, and common table expressions. Metrics were selected to match each business question.

Metric design

- Revenue = order quantity × unit price.
- Production efficiency was assessed using production cost per kilogram alongside production volume and batch-level cost.
- Supplier defect rate used unique supplier–batch associations so the same supplier/batch was not counted multiple times when multiple raw materials were involved.
- Inventory movement was evaluated using inbound, outbound, total movement, and net inventory change.
- Monthly performance was evaluated using both order volume and revenue.

Visualization

Excel was used as a stepping stone for chart creation and dashboard layout. A Power BI dashboard was subsequently built to provide a more interactive version of the analysis.

Scope refinement

Two additional questions were evaluated but excluded from the final analysis because the dataset provided little differentiation: customers purchased similar numbers of distinct products, and every order had a one-day order-to-ship interval. Excluding them kept the final project focused on analyses with stronger business value.

5.1 Business Question 1: Which products generate the highest revenue?

Why It Matters

Understanding which products contribute the most revenue helps management prioritize production capacity, inventory planning, and sales efforts.

SQL Used

```
SELECT p.product_name,  
ROUND(SUM(oi.quantity_kg * oi.unit_price), 2) AS revenue_per_product  
FROM order_items AS oi  
JOIN products AS p ON oi.product_id = p.product_id  
GROUP BY p.product_id, p.product_name  
ORDER BY revenue_per_product DESC;
```

Findings

Revenue is relatively evenly distributed across the product portfolio, with no single product dominating overall sales.

Business Insight

Because revenue is distributed relatively evenly, the company should maintain a balanced production and inventory strategy rather than concentrating resources on only a few products.

Recommendation

Use profit margins, production costs, product mix, and customer demand in future analysis to identify products that offer the greatest business value beyond revenue alone.

5.2 Business Question 2: Which customers generate the highest revenue?

Why It Matters

Identifying the highest-value customers helps management understand where revenue is concentrated and where retention efforts may have the greatest impact.

SQL Used

```
SELECT c.customer_id, c.company_name, c.industry, c.country,
ROUND(SUM(oi.quantity_kg * oi.unit_price), 2) AS total_revenue,
DENSE_RANK() OVER (ORDER BY SUM(oi.quantity_kg * oi.unit_price) DESC) AS revenue_rank,
COUNT(DISTINCT oi.order_id) AS number_of_orders,
ROUND(SUM(oi.quantity_kg * oi.unit_price) /
COUNT(DISTINCT oi.order_id), 2) AS AOV,
DENSE_RANK() OVER (ORDER BY ROUND(SUM(oi.quantity_kg * oi.unit_price) /
COUNT(DISTINCT oi.order_id), 2) DESC) AS AOV_rank
FROM order_items AS oi
JOIN orders AS o ON oi.order_id = o.order_id
JOIN customers AS c ON o.customer_id = c.customer_id
GROUP BY c.customer_id, c.company_name, c.industry, c.country
ORDER BY total_revenue DESC;
```

Findings

The top five customers generated the highest total revenue but did not rank among the top five in Average Order Value. They placed substantially more orders, indicating that revenue was driven primarily by purchasing frequency.

Business Insight

Customer value appears to be driven more by repeat purchasing behavior than by exceptionally large orders.

Recommendation

Focus retention efforts on high-frequency customers through account management, loyalty initiatives, and proactive engagement.

5.3 Business Question 3: How do monthly sales and order volumes change over time?

Why It Matters

Monitoring monthly order volume and revenue helps management identify stronger and weaker periods and potential seasonal patterns.

SQL Used

```
SELECT MONTH(order_date) AS month_orders,  
COUNT(DISTINCT o.order_id) AS number_of_orders,  
DENSE_RANK() OVER (ORDER BY COUNT(DISTINCT o.order_id) DESC) AS monthly_orders_rank,  
ROUND(SUM(oi.quantity_kg * oi.unit_price), 2) AS total_revenue,  
DENSE_RANK() OVER (ORDER BY SUM(oi.quantity_kg * oi.unit_price) DESC) AS revenue_rank  
FROM orders AS o  
LEFT JOIN order_items AS oi ON o.order_id = oi.order_id  
GROUP BY MONTH(order_date);
```

Findings

October recorded both the highest number of orders and the highest total revenue. September and April recorded the fewest orders and the lowest revenue in the dataset.

Business Insight

Monthly revenue appears strongly associated with order volume: higher customer activity corresponded with higher sales revenue.

Recommendation

Monitor lower-volume months. If similar patterns persist over longer periods, investigate operational or market factors and consider targeted sales or marketing initiatives.

5.4 Business Question 4: Which manufacturing plants have the highest production costs?

Why It Matters

Production cost is a major driver of profitability. Comparing plants can help management investigate differences in efficiency, resource use, equipment utilization, and operating practices.

SQL Used

```
SELECT p.plant_id, p.plant_name,
ROUND(SUM(production_cost), 2) AS total_production_cost,
COUNT(batch_id) AS production_batches,
ROUND(AVG(production_cost), 2) AS avg_production_cost,
ROUND(SUM(quantity_produced_kg), 2) AS total_quantity_produced_kg,
ROUND(SUM(production_cost) / SUM(quantity_produced_kg), 2) AS production_cost_per_kg
FROM plants AS p
JOIN production_batches AS pb ON p.plant_id = pb.plant_id
GROUP BY p.plant_id, p.plant_name
ORDER BY total_production_cost DESC;
```

Findings

Batangas incurred the highest total production cost at ₱8.96 million and the highest average production cost per batch at ₱30,176.87. It produced 832,822.60 kg at ₱10.76/kg. Laguna processed 324 batches, produced 876,808.26 kg, and achieved the lowest average batch cost (₱26,775.55) and lowest cost per kilogram (₱9.89). Cebu had the lowest total production cost at ₱7.80 million and 279 batches, with ₱10.02/kg.

Business Insight

Laguna demonstrates the strongest cost efficiency because it combines the highest production volume with the lowest cost per kilogram. Batangas has the highest cost per kilogram despite producing less than Laguna.

Recommendation

Use Laguna as a benchmark. Review Batangas operating practices, equipment utilization, resource consumption, and workflows. Future analysis should incorporate product mix, batch complexity, and equipment utilization.

5.5 Business Question 5: Which suppliers are associated with the highest quality defect rates?

Why It Matters

Supplier-associated defect rates provide an initial quality-performance signal that can support supplier monitoring and further investigation.

SQL Used

```
WITH supplier_batch_results AS (  
  SELECT DISTINCT rm.supplier_id, qt.batch_id, qt.passed  
  FROM quality_tests AS qt  
  JOIN batch_material_usage AS bmu ON qt.batch_id = bmu.batch_id  
  JOIN raw_materials AS rm ON bmu.material_id = rm.material_id  
)  
SELECT s.supplier_name,  
  SUM(CASE WHEN sbr.passed = 0 THEN 1 ELSE 0 END) AS failed_associated_batches,  
  COUNT(*) AS total_associated_batches,  
  ROUND(SUM(CASE WHEN sbr.passed = 0 THEN 1 ELSE 0 END) /  
    COUNT(*) * 100, 2) AS defect_rate  
FROM supplier_batch_results AS sbr  
JOIN suppliers AS s ON sbr.supplier_id = s.supplier_id  
GROUP BY s.supplier_id, s.supplier_name  
ORDER BY defect_rate DESC;
```

Findings

Nexus Raw Materials recorded the highest supplier-associated defect rate at 7.14%, followed by Orion Industrial Supply at 7.02% and Apex Chemical Supply at 6.95%. Atlas Industrial Chemicals had the lowest rate at 1.16%. Global Solvent Traders was associated with 304 batches but had a relatively low 4.93% defect rate.

Business Insight

Supplier quality was generally consistent, with defect rates ranging from 1.16% to 7.14%. The narrow spread suggests that no supplier exhibited exceptionally poor performance and that quality issues may also depend on material or production factors.

Recommendation

Continue monitoring higher-rate suppliers, particularly Nexus, Orion, and Apex, while considering batch volume. Future analysis should include material type, defect category, and production conditions.

5.6 Business Question 6: Which products experience the highest inventory movement?

Why It Matters

Inventory movement indicates operational activity and helps management evaluate whether inbound supply is keeping pace with outbound demand.

SQL Used

```
SELECT it.product_id, p.product_name,  
SUM(CASE WHEN it.transaction_type = 'IN' THEN it.quantity_kg ELSE 0 END) AS total_in,  
SUM(CASE WHEN it.transaction_type = 'OUT' THEN it.quantity_kg ELSE 0 END) AS total_out,  
SUM(it.quantity_kg) AS total_movement,  
SUM(CASE WHEN it.transaction_type = 'IN' THEN it.quantity_kg ELSE 0 END)  
- SUM(CASE WHEN it.transaction_type = 'OUT' THEN it.quantity_kg ELSE 0 END)  
AS net_inventory_change  
FROM inventory_transactions AS it  
JOIN products AS p ON it.product_id = p.product_id  
GROUP BY it.product_id, p.product_name  
ORDER BY total_movement DESC;
```

Findings

Slip Additive SA-12 recorded the highest movement at 186,829.51 kg, followed by Copper Sulfate Pentahydrate at 184,539.61 kg and Acetone at 181,549.99 kg. Acetone (-36,228.10 kg), Aluminum Sulfate (-28,829.45 kg), and Slip Additive SA-12 (-26,557.88 kg) had notable net decreases. Sodium Hypochlorite (+42,147.57 kg) and Sulfuric Acid 98% (+29,167.42 kg) had the largest net increases.

Business Insight

High movement alone does not indicate effective inventory management. Net inventory change identifies products that may be depleting faster than replenishment as well as products that may be accumulating inventory.

Recommendation

Monitor high-movement products with negative net changes to reduce stockout risk. Review large positive changes to ensure inventory aligns with expected demand and to prevent unnecessary buildup.

6. Final Business Summary

Key strengths

- Balanced revenue contribution across products.
- High-value customer revenue supported by repeat purchasing.
- Laguna combines production scale with strong cost efficiency.
- Supplier defect rates are broadly consistent.

Operational issues and opportunities

- Batangas has the highest production cost per kilogram.
- Some high-movement products show substantial negative net inventory changes.
- Some products show large positive net inventory changes that warrant demand review.
- Supplier defect rates should not be interpreted without material and production context.

Priority recommendations

- Benchmark plant operations against Laguna and investigate Batangas unit costs.
- Use purchase frequency as a customer-retention metric.
- Monitor weaker sales months and investigate recurring demand patterns.
- Monitor supplier quality using defect rate together with batch volume and material context.
- Track total inventory movement together with net inventory change.

Analytical limitations

The findings are descriptive and based on the available simulated dataset. They do not establish causality. Revenue is not profit; production comparisons do not control for product mix or batch complexity; supplier defect rates may be affected by material and production conditions; and monthly patterns cannot be treated as confirmed seasonality without a longer time series.

7. Project Deliverables

Deliverable	Purpose
README.md	Project overview, methodology, schema, findings, and repository guidance
ERD	Visual representation of relational structure and relationships
Database creation SQL	Creates the relational schema
Data validation SQL	Documents data-quality and integrity checks
EDA SQL	Documents exploratory analysis
Business questions SQL	Final SQL used to answer selected questions
Excel dashboard	Initial visualization and Excel practice
Power BI dashboard	Interactive final visualization
Project report	Consolidated findings, insights, recommendations, and methodology

Closing statement

The project demonstrates an end-to-end approach to turning relational business data into actionable insights. Its central strength is the analytical workflow: validate the data, define an appropriate metric, query the database, challenge the result when necessary, communicate the finding visually, and translate it into a practical recommendation.